

透過 Private Service Connect 存取多區域 MongoDB Atlas

來源：<https://codelabs.developers.google.com/codelabs/psc-mongo-globalaccess?hl=zh-tw>

步驟數：13

在這個程式碼研究室中，您將瞭解如何透過全域存取權設定及驗證 Private Service Connect 對 MongoDB 的存取權。

目錄

1. [1. 簡介](#)
2. [2. 啟用全域存取權](#)
3. [3. 建構項目](#)
4. [4. 事前準備](#)
5. [5. 消費者設定](#)
6. [6. 建立多區域 MongoDB Atlas 叢集](#)
7. [7. 為 us-west1 建立私人端點](#)
8. [8. 為 us-west2 建立私人端點](#)
9. [9. 從私人端點連線至 MongoDB Atlas](#)
10. [10. 安裝 mongosh 應用程式](#)
11. [11. 容錯移轉使用中的 MongoDB 區域 \(us-west1\)](#)
12. [12. 清除](#)
13. [13. 恭喜](#)

程式碼研究室簡介

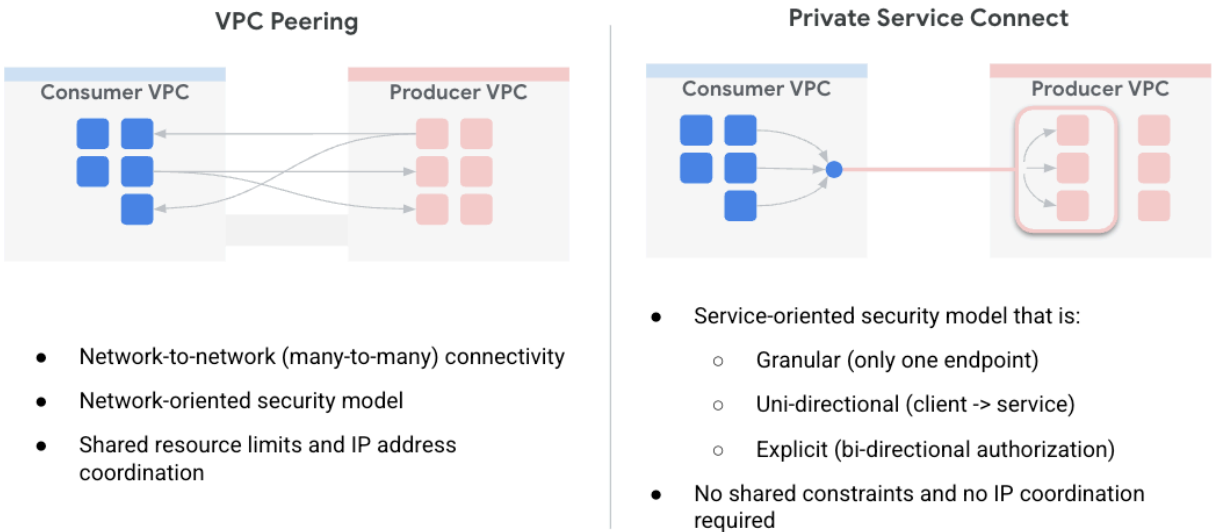
subject上次更新時間：3月 27, 2026

account_circle作者：Deepak Michael

1. 簡介

Private Service Connect (PSC) 可簡化安全且私密地使用服務的方式。這個模型可讓服務用戶在不同機構之間私下連線至服務供應商，並免除虛擬私有雲對接的需求，大幅簡化網路架構。圖 1 說明 VPC 對等互連和 PSC 屬性。

圖 1.



服務用戶可彈性選擇如何將私人 IP 位址分配給服務，同時免除管理生產者虛擬私有雲子網路範圍的負擔。現在只要使用服務連結，即可將虛擬私有雲中選取的虛擬 IP 指派給這類服務。

在本程式碼研究室中，您將建構完整的 Private Service Connect 架構，說明如何搭配 MongoDB Atlas 使用 PSC 全域存取權。

全域存取權可讓用戶端跨區域界線連線至 Private Service Connect (PSC)。這項功能有助於在多個區域代管的受管理服務中建立高可用性，或允許用戶端存取與用戶端不在同一區域的服務。

2. 啟用全域存取權

全域存取權是選用功能，可在取用端轉送規則中設定。以下指令顯示設定方式：

```
gcloud beta compute forwarding-rules create psc-west \  
  --region=us-west1 \  
  --network=consumer-vpc \  
  --address=psc-west-address \  
  --target-service-attachment=projects/.../serviceAttachments/sa-west \  
  --allow-psc-global-access
```

- `--allow-psc-global-access` 旗標可啟用 Private Service Connect 端點的全域存取權
- 啟用全域存取權後，用戶端可以位於與 Private Service Connect 轉送規則不同的區域，但轉送規則仍須與所連線的服務連結位於相同區域。
- 如要啟用全域存取權，生產者服務附件不需要任何設定。這純粹是消費者端的選項。

您也可以隨時為現有端點開啟或關閉全域存取權。在現有端點上啟用全域存取權時，有效連線不會中斷。使用下列指令，在現有的轉送規則上啟用全域存取權：

```
gcloud beta compute forwarding-rules update psc-west --allow-psc-global-access
```

停用全域存取權

您也可以使用 `--no-allow-psc-global-access` 旗標，在現有的轉送規則中停用全域存取權。請注意，執行這項指令後，所有有效的跨區域流量都會終止。

```
gcloud beta compute forwarding-rules update psc-west --no-allow-psc-global-access
```

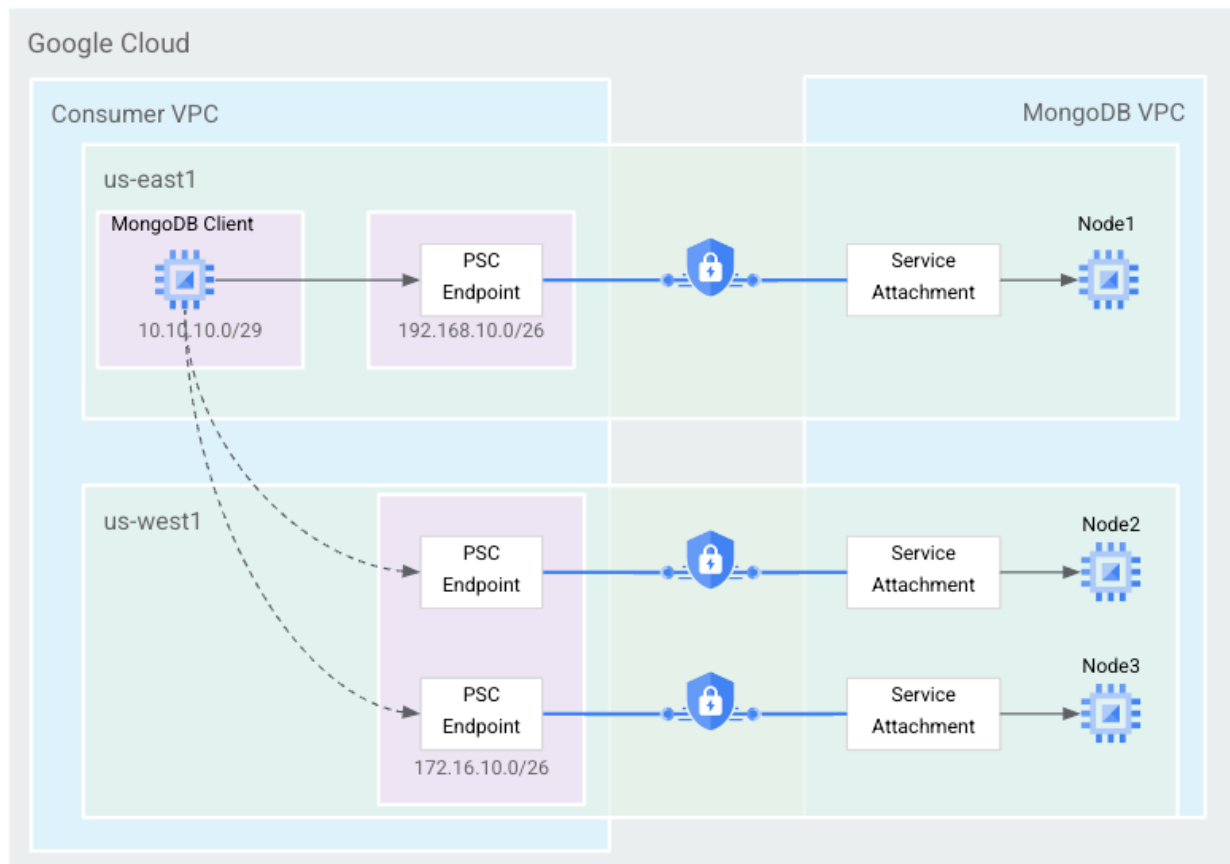
注意：全域存取權是預先發布版功能，必須使用「gcloud beta」才能啟用。

3. 建構項目

1. 系統會建立多區域 MongoDB Atlas 叢集 (拓撲如圖 2 所示)，其中 us-west1 和 us-west2 區域分別有一個和兩個節點。
2. 用戶虛擬私有雲和相關聯的 VM，可存取 us-west1 和 us-west2 中的 MongoDB 叢集。
3. 虛擬私有雲，以及分別位於 us-west1 和 us-west2 區域的兩個子網路，每個子網路至少有 64 個可用 IP 位址 (建立子網路時請使用 /26 以下的範圍)。

MongoDB 用戶端會安裝在消費者 VPC 的 vm1 上。如果 us-west1 中的主要節點發生故障，用戶端就能透過 us-west2 中的新主要節點讀取/寫入資料。

圖 2.



附註：Codelab 會根據圖示拓撲提供設定和驗證步驟，請視需要修改程序，以符合貴機構的需求。本程式碼研究室不包含 IAM 權限。

課程內容

- 如何在兩個區域中建立及部署虛擬私有雲和子網路
- 如何部署多區域 MongoDB Atlas 叢集
- 如何建立私人端點
- 如何連線至 MongoDB
- 如何執行及驗證多區域 MongoDB 容錯移轉

軟硬體需求

- Google Cloud 專案
- 每個區域提供一個 /26 子網路
- 專案擁有者或機構擁有者必須有權存取 MongoDB Atlas，才能建立叢集層級為 M10 以上的 MongoDB 叢集。(請使用 GETATLAS 取得免費抵免額，以執行 PoV)

注意：私人端點僅適用於專屬叢集，也就是 M10 以上的叢集。

步驟 4 · 約 2 分鐘

4. 事前準備

更新專案以支援程式碼研究室

本程式碼研究室會使用 `$variables`，協助您在 Cloud Shell 中實作 gcloud 設定。

在 Cloud Shell 中執行下列操作：

```
gcloud config list project
gcloud config set project [YOUR-PROJECT-NAME]
projectname=YOUR-PROJECT-NAME
echo $projectname
```

步驟 5 · 約 10 分鐘

5. 消費者設定

建立消費者虛擬私有雲

在 Cloud Shell 中執行下列操作：

```
gcloud compute networks create consumer-vpc --project=$projectname --subnet-mode=custom
```

建立消費者子網路

在 Cloud Shell 中執行下列操作：

```
gcloud compute networks subnets create vm-subnet --project=$projectname --range=10.10.10.0/29 --network=consumer-vpc --region=us-west1
```

在 Cloud Shell 中，為 us-west1 建立消費者端點：

```
gcloud compute networks subnets create psc-endpoint-us-west1 --project=$projectname --range=192.168.10.0/26 --network=consumer-vpc --region=us-west1
```

在 Cloud Shell 中，為 us-west2 建立消費者端點：

```
gcloud compute networks subnets create psc-endpoint-us-west2 --project=$projectname --range=172.16.10.0/26 --network=consumer-vpc --region=us-west2
```

Cloud Router 和 NAT 設定

由於 VM 執行個體沒有外部 IP 位址，因此本程式碼研究室使用 Cloud NAT 安裝軟體套件。

在 Cloud Shell 中建立 Cloud Router。

```
gcloud compute routers create consumer-cr --network consumer-vpc --region us-west1
```

在 Cloud Shell 中建立 NAT 閘道。

```
gcloud compute routers nats create consumer-nat --router=consumer-cr --auto-allocate-nat-external-ips --nat-all-subnet-ip-ranges --region us-west1
```

執行個體 vm1 設定

在下一節中，您將建立 Compute Engine 執行個體 vm1。

在 Cloud Shell 中建立執行個體 vm1。


```
gcloud compute instances create vm1 \  
  --project=$projectname \  
  --zone=us-west1-a \  
  --machine-type=e2-micro \  
  --network-interface=subnet=vm-subnet,no-address \  
  --maintenance-policy=MIGRATE \  
  --provisioning-model=STANDARD \  
--create-disk=auto-delete=yes,boot=yes,device-name=vm1,image=projects/ubuntu-os-  
cloud/global/images/ubuntu-2004-focal-  
v20230213,mode=rw,size=10,type=projects/$projectname/zones/us-west1-a/diskTypes/pd-balanced  
  --metadata startup-script="#! /bin/bash  
    sudo apt-get update  
    sudo apt-get install tcpdump -y  
    sudo apt-get install dnsutils -y"
```

如要允許 IAP 連線至您的 VM 執行個體，請根據以下條件建立防火牆規則：

- 套用至所有您希望能透過 IAP 存取的 VM 執行個體。
- 允許來自 IP 範圍 35.235.240.0/20 的輸入流量。這個範圍包含 IAP 用於 TCP 轉送的所有 IP 位址。

在 Cloud Shell 中，建立 IAP 防火牆規則。

```
gcloud compute firewall-rules create ssh-iap-consumer-vpc \  
  --network consumer-vpc \  
  --allow tcp:22 \  
  --source-ranges=35.235.240.0/20
```

6. 建立多區域 MongoDB Atlas 叢集

- 開始設定 PSC 前，我們需要先設定 Atlas 叢集。您可以透過下列兩種方式訂閱 MongoDB Atlas：
- 如果您有 Google Cloud 帳戶，可以透過 [Google Cloud Marketplace](#) 購買。如要設定訂閱方案，請參閱[說明文件](#)。
- 使用 [Atlas 註冊頁面](#)。
- 訂閱 Atlas 後，按一下「Build a Database」按鈕，如下所示。



Create a database

Choose your cloud provider, region, and specs.

Build a Database

Once your database is up and running, live migrate an existing MongoDB database into Atlas with our [Live Migration Service](#).

注意：如果您是 Atlas 訂閱者，請選取「建立」來部署新資料庫

Database Deployments

Find a database deployment...

+ Create

- 建立新叢集 → 專屬

CLUSTERS > CREATE NEW CLUSTER

Create New Cluster

Serverless

Dedicated

Shared

For production applications with sophisticated workload requirements. Advanced configuration controls.

Network isolation, end-to-end encryption, and fine-grained access controls.

On-demand performance advice, including index and schema suggestions.

- 雲端服務供應商和區域 → Google Cloud
- 多雲端、多區域和工作負載隔離 → 已選取 (藍色勾號)

Cloud Provider & Region

GCP
5 Electable Multi-Region Nodes



Multi-Cloud, Multi-Region & Workload Isolation (M10+ clusters)

Distribute data across clouds or regions for improved availability and local read performance, or introduce read-only and analytics nodes. [Learn more](#)



As configured below, your cluster is:

- ✓ Available during partial region outage
- ✗ Not available during full region outage
- ✗ Not available during a cloud provider outage

- 可選取的節點 → us-west1 (1 個節點)、us-west2 (2 個節點)

Electable nodes for high availability

Configure 3, 5, or 7 nodes across multiple regions to better withstand data center outages

★ Recommended region ⓘ

Provider	Region	Priority	Nodes	Action
GCP ▾	Oregon (us-west1) ★ ▾	HIGHEST	1	
GCP ▾	Los Angeles (us-west2) ★ ▾		2	
+ Add a provider/region				
Total: 3 Electable Nodes				

Read-only nodes for optimal local reads

Add replicas in additional regions to optimize for local reads in any of your service areas

Provider	Region	Nodes	Action
+ Add a provider/region			
Total: 0			

Analytics nodes for workload isolation

Isolate analytics queries on nodes that will not contend with your operational workload

Provider	Region	Nodes	Action
+ Add a provider/region			
Total: 0			

- 叢集層級 → M10，其他設定保留預設值

Cluster Tier

M10 (1.7 GB RAM, 40 GB Storage)
2,400 IOPS, Encrypted, Auto-expand Storage

Hourly price is for a MongoDB replica set with 3 data bearing servers.

Dedicated Clusters for development environments and low-traffic applications

Tier	RAM	Storage	vCPU	Price
✓ M10	1.7 GB	10 GB	0.5 vCPUs	from \$0.08/hr
Storage 10 GB is included in the base price.				
10 GB  128 GB 40 GB				
Auto-scale ☒ Cluster Tier Scaling View docs				
Minimum cluster size M10 Maximum cluster size M40				
☒ Allow cluster to be scaled down				
☒ Storage Scaling ⓘ				
IOPS 2,400 Total IOPS up to 1,200 Read IOPS and 1,200 Write IOPS				
Additional Info 1500 max connections Low to Moderate network performance				
M20	3.75 GB	20 GB	1 vCPU	from \$0.19/hr

- 叢集名稱 → psc-mongodb-uswest1-uswest2

Cluster Name

psc-mongodb-uswest1-uswest2

One time only: once your cluster is created, you won't be able to change its name.

psc-mongodb-uswest1-uswest2

Cluster names can only contain ASCII letters, numbers, and hyphens.

- 選取「→」→「建立叢集」

Cancel

Create Cluster

- 建立資料庫需要 7 到 10 分鐘

[psc-mongodb-uswest1-uswest2](#) [Connect](#) [View Monitoring](#) [Browse Collections](#) [...](#)

 Your cluster is being created...
New clusters take between 7-10 minutes to provision.

VERSION	PREFERRED REGION	CLUSTER TIER	TYPE	BACKUPS	LINKED APP SERVICES	BI CONNECTOR	ONLINE ARCHIVE	ATLAS SEARCH
5.0.14	GCP / Oregon (us-west1)	M10 (General)	Multi Region Replica Set - 3 nodes	Active	None Linked	Disabled	None	Create Index

部署叢集後的檢視畫面

[psc-mongodb-uswest1-uswest2](#) [Connect](#) [View Monitoring](#) [Browse Collections](#) [...](#)

Enhance Your Experience
For high performance production applications, upgrade to a M30 cluster now!
[Dismiss](#) [Upgrade](#)

R 1.1
W 0.03
Last 9 seconds
1.1/s

Connections 37.0
Last 9 seconds
37.0

In 3.8 KB/s
Out 27.7 KB/s
Last 9 seconds
28.2 KB/s

Disk Usage 1.5 GB
Last 6 minutes
10.0 GB

VERSION	PREFERRED REGION	CLUSTER TIER	TYPE	BACKUPS	LINKED APP SERVICES	BI CONNECTOR	ONLINE ARCHIVE	ATLAS SEARCH
5.0.15	GCP / Oregon (us-west1)	M10 (General)	Multi Region Replica Set - 3 nodes	Active	None Linked	Disabled	None	Create Index

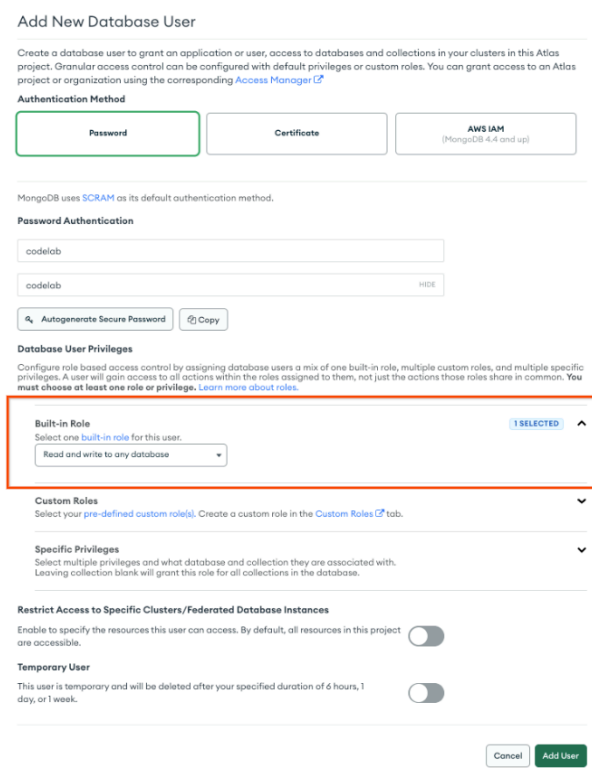
步驟 7 · 約 10 分鐘

7. 為 us-west1 建立私人端點

- 登入 Atlas 帳戶並前往專案。

建立新使用者，允許讀取/寫入任何資料庫

依序選取「安全性」→「資料庫存取權」，然後新增資料庫使用者。以下為範例，使用者名稱和密碼設定為 codelab。請務必選取內建角色，以便讀取及寫入任何資料庫。



Add New Database User

Create a database user to grant an application or user, access to databases and collections in your clusters in this Atlas project. Granular access control can be configured with default privileges or custom roles. You can grant access to an Atlas project or organization using the corresponding [Access Manager](#).

Authentication Method

☒ Password ☐ Certificate ☐ AWS IAM (MongoDB 4.4 and up)

MongoDB uses SCRAM as its default authentication method.

Password Authentication

HIDE

Database User Privileges

Configure role based access control by assigning database users a mix of one built-in role, multiple custom roles, and multiple specific privileges. A user will gain access to all actions within the roles assigned to them, not just the actions those roles share in common. You must choose at least one role or privilege. [Learn more about roles.](#)

Built-in Role
Select one built-in role for this user. 1 SELECTED

Custom Roles
Select your pre-defined custom role(s). Create a custom role in the Custom Roles tab.

Specific Privileges
Select multiple privileges and what database and collection they are associated with. Leaving collection blank will grant this role for all collections in the database.

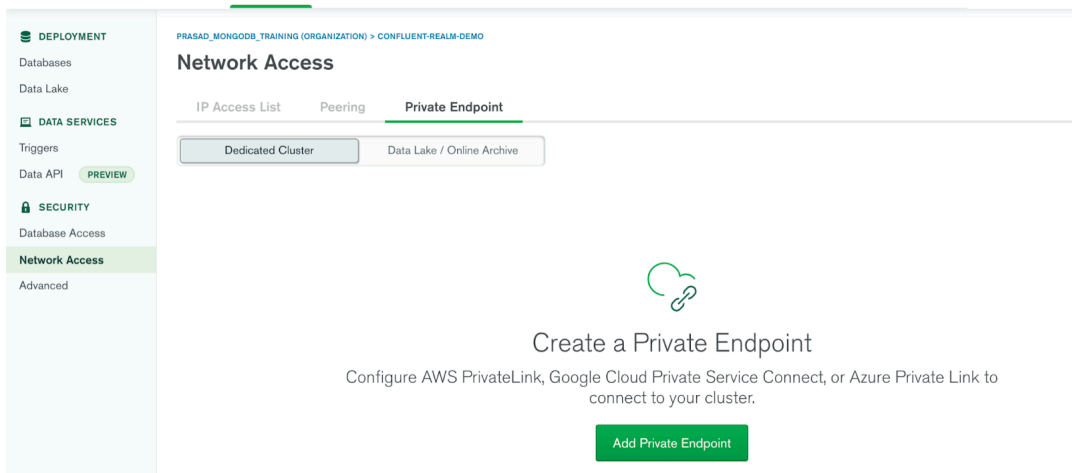
Restrict Access to Specific Clusters/Federated Database Instances
Enable to specify the resources this user can access. By default, all resources in this project are accessible. ☐

Temporary User
This user is temporary and will be deleted after your specified duration of 6 hours, 1 day, or 1 week. ☐

- 在「安全性」→「網路存取」下方，「IP 存取清單」不需要任何項目

在 MongoDB Atlas 中準備私人端點

- 依序選取「網路存取」→「私人端點」→「專屬叢集」→「新增私人端點」



DEPLOYMENT

Databases
Data Lake

DATA SERVICES

Triggers
Data API PREVIEW

SECURITY

Database Access
Network Access
Advanced

PRASAD_MONGODB_TRAINING (ORGANIZATION) > CONFLUENT-REALM-DEMO

Network Access

IP Access List Peering **Private Endpoint**



Create a Private Endpoint

Configure AWS PrivateLink, Google Cloud Private Service Connect, or Azure Private Link to connect to your cluster.

注意：每個專案的每個區域僅支援一個私人端點。如果 us-west1 和 us-west2 中有私人端點，請刪除叢集，然後在其他區域重新部署。

雲端服務供應商

- 選取 Google Cloud，然後按一下「下一步」

Create a Private Endpoint M10 & ABOVE ONLY

Cloud Provider

Service Attachments

Endpoints

Set Up Endpoints

Finalize







Close

Next

服務連結

- 選取 us-west1 地區，然後按一下「下一步」

Create a Private Endpoint M10 & ABOVE ONLY

✓ Cloud Provider

Service Attachments

Endpoints

Set Up Endpoints

Finalize

 The process of creating a private endpoint in Google Cloud includes the creation of [Private Service Connect endpoints](#) that will support growing Atlas Usage. This will enable you to connect to multiple Atlas clusters privately. [See documentation to learn more](#)

Atlas Region

Select Atlas Region

★ Recommended region 

 Oregon (us-west1) ★

Billing

Pay-as-you-go! After this step, you will be billed hourly for each Google Cloud Private Service Connect region at a rate of \$0.03 per GB.

Back

Next

端點

- 如要建立 Private Service Connect 端點，請提供下列資訊：
- Google Cloud 雲端專案 ID：選取「顯示操作說明」查看詳細資料
- 虛擬私有雲名稱：consumer-vpc
- 子網路名稱：psc-endpoint-us-west1
- Private Service Connect 端點前置字元：psc-endpoint-us-west1

Create a Private Endpoint M10 & ABOVE ONLY

✓ Cloud Provider ✓ Service Attachments **Endpoints** Set Up Endpoints Finalize

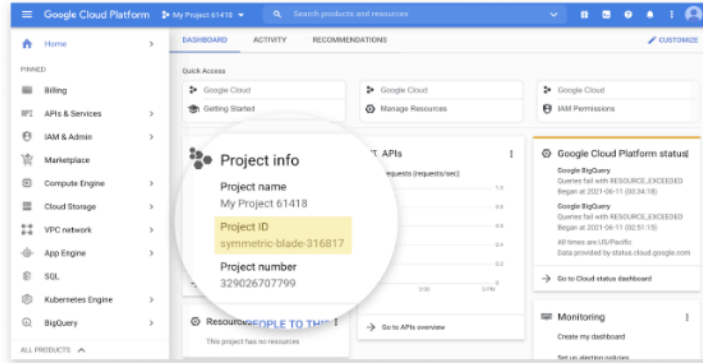
✓ Atlas Endpoint Service Ready

Your Atlas Endpoint Service is ready to accept private endpoint connections. Create your Private Service Connect endpoints in your Google Cloud project using the [Google Cloud CLI](#). When you create Private Service Connect endpoints, you specify a subnet in your VPC network; you may either create a new subnet to encapsulate your endpoints or use an existing one. Enter the following information to populate the script in the next step.

Google Cloud Project ID

Find this in the Dashboard page in the Google Cloud Platform.
[Hide instruction](#)

① Navigate to the Google Cloud Platform Home Dashboard and locate the Project ID.



VPC Name

Find this on the VPC Networks page in the Google Cloud Platform.
[Show instruction](#)

Subnet Name ①

Find this on the VPC Networks page in Google Cloud Platform. This subnet will need to have at least 50 unassigned IP addresses.
[Show instruction](#)

Private Service Connect Endpoint Prefix

Enter a unique name for your Private Endpoint. This will be the prefix for all forwarding rules created.

Back

Next

注意：端點建立作業需要 3 到 4 分鐘，完成後即可選取「下一步」

Create a Private Endpoint M10 & ABOVE ONLY

✓ Cloud Provider ✓ Service Attachments **Endpoints** Set Up Endpoints Finalize

✓ Creating Atlas Endpoint Service... This might take 3-4 minutes to process

設定端點

在下一節中，系統會產生名為 setup_psc.sh 的殼層指令碼，請將其儲存在本機。儲存後，請編輯殼層指令碼，允許 PSC 全域存取。您可以在 Google Cloud 雲端專案的 Cloud Shell 中執行這項操作。

- 範例殼層指令碼，輸出內容的值會有所不同

Create a Private Endpoint M10 & ABOVE ONLY

✓ Cloud Provider ✓ Service Attachments ✓ Endpoints **Set Up Endpoints** Finalize

Use the following generated script to set up your private endpoints. This script will reserve static internal IP addresses in the subnet specified in the previous step, and create forwarding rules to the Atlas Private Endpoint Service. For alternative options to set up your Private Service Connect endpoints, please review our [documentation](#).

1. Install gcloud command-line interface

You will need to have the gcloud command-line interface installed for the next steps. Follow the [instructions here](#) to install and run it on your local machine.

2. Copy and save shell script

Copy this script and save it to a file named `setup_psc.sh`.

```
#!/bin/bash
gcloud config set project [REDACTED]

for i in {0..49}
do
  gcloud compute addresses create psc-endpoint-us-west1-ip-$i --region=us-west1 --
  subnet=psc-endpoint-us-west1
done

for i in {0..49}
do
  if [ $(gcloud compute addresses describe psc-endpoint-us-west1-ip-$i --region=us
  -west1 --format="value(status)") != "RESERVED" ]; then
    echo "psc-endpoint-us-west1-ip-$i is not RESERVED";
    exit 1;
  fi
done

for i in {0..49}
do
  gcloud compute forwarding-rules create psc-endpoint-us-west1-$i --region=us-west
  1 --network=consumer-vpc --address=psc-endpoint-us-west1-ip-$i --target-service-at
  tachment=projects/p-npwszmzelxznmaejhj2vn1q0q/regions/us-west1/serviceAttachments/s
  a-us-west1-61485ec2ae9d2e48568bf84f-$i
done

if [ $(gcloud compute forwarding-rules list --regions=us-west1 --format="csv[no-he
  ading](name)" --filter="(name:psc-endpoint-us-west1*)" | wc -l) -gt 50 ]; then
  echo "Project has too many forwarding rules that match prefix psc-endpoint-us-we
  st1. Either delete the competing resources or choose another endpoint prefix."
  exit 2;
fi

gcloud compute forwarding-rules list --regions=us-west1 --format="json(IPAddress,n
  ame)" --filter="name:(psc-endpoint-us-west1*)" > atlasEndpoints-psc-endpoint-us-we
  st1.json
```

3. Run shell script

Run the script in your shell. Navigate to the directory where the script is saved and run it using the command below. This will output a file called `atlasEndpoints-psc-endpoint-us-west1.json` containing a list of IP addresses and forwarding rule names that you will need in the next step.

sh setup_psc.sh

 Copy

Back

Next

- 從 MongoDB 控制台複製殼層指令碼，然後將內容儲存在 Google Cloud Cloud Shell 終端機中，並確保將指令碼儲存為 `setup_psc.sh`

更新前的範例：


```
#!/bin/bash
gcloud config set project yourprojectname

for i in {0..49}
do
    gcloud compute addresses create psc-endpoint-us-west1-ip-$i --region=us-west1 --subnet=psc-
endpoint-us-west1
done

for i in {0..49}
do
    if [ $(gcloud compute addresses describe psc-endpoint-us-west1-ip-$i --region=us-west1 --
format="value(status)") != "RESERVED" ]; then
        echo "psc-endpoint-us-west1-ip-$i is not RESERVED";
        exit 1;
    fi
done

for i in {0..49}
do
    gcloud compute forwarding-rules create psc-endpoint-us-west1-$i --region=us-west1 --
network=consumer-vpc --address=psc-endpoint-us-west1-ip-$i --target-service-
attachment=projects/p-npwszmzelxznmaejhj2vnlq0q/regions/us-west1/serviceAttachments/sa-us-
west1-61485ec2ae9d2e48568bf84f-$i
done

if [ $(gcloud compute forwarding-rules list --regions=us-west1 --format="csv[no-heading]
(name)" --filter="(name:psc-endpoint-us-west1*)" | wc -l) -gt 50 ]; then
    echo "Project has too many forwarding rules that match prefix psc-endpoint-us-west1. Either
delete the competing resources or choose another endpoint prefix."
    exit 2;
fi

gcloud compute forwarding-rules list --regions=us-west1 --format="json(IPAddress,name)" --
filter="name:(psc-endpoint-us-west1*)" > atlasEndpoints-psc-endpoint-us-west1.json
```

更新殼層指令碼，支援全域存取

使用 nano 或 vi 編輯器找出並更新殼層指令碼，語法如下：

```
gcloud beta compute forwarding-rules create psc-endpoint-us-west1-$i --region=us-west1 --
network=consumer-vpc --address=psc-endpoint-us-west1-ip-$i --target-service-
attachment=projects/p-npwszmzelxznmaejhj2vnlq0q/regions/us-west1/serviceAttachments/sa-us-
west1-61485ec2ae9d2e48568bf84f-$i --allow-psc-global-access
```

更新後的範例：

```
#!/bin/bash
gcloud config set project yourprojectname

for i in {0..49}
do
    gcloud compute addresses create psc-endpoint-us-west1-ip-$i --region=us-west1 --subnet=psc-endpoint-us-west1
done

for i in {0..49}
do
    if [ $(gcloud compute addresses describe psc-endpoint-us-west1-ip-$i --region=us-west1 --format="value(status)") != "RESERVED" ]; then
        echo "psc-endpoint-us-west1-ip-$i is not RESERVED";
        exit 1;
    fi
done

for i in {0..49}
do
    gcloud beta compute forwarding-rules create psc-endpoint-us-west1-$i --region=us-west1 --network=consumer-vpc --address=psc-endpoint-us-west1-ip-$i --target-service-attachment=projects/p-npwszmzelxznmaejhj2vnlq0q/regions/us-west1/serviceAttachments/sa-us-west1-61485ec2ae9d2e48568bf84f-$i --allow-psc-global-access
done

if [ $(gcloud compute forwarding-rules list --regions=us-west1 --format="csv[no-heading](name)" --filter="(name:psc-endpoint-us-west1*)" | wc -l) -gt 50 ]; then
    echo "Project has too many forwarding rules that match prefix psc-endpoint-us-west1. Either delete the competing resources or choose another endpoint prefix."
    exit 2;
fi

gcloud compute forwarding-rules list --regions=us-west1 --format="json(IPAddress,name)" --filter="name:(psc-endpoint-us-west1*)" > atlasEndpoints-psc-endpoint-us-west1.json
```

執行殼層指令碼

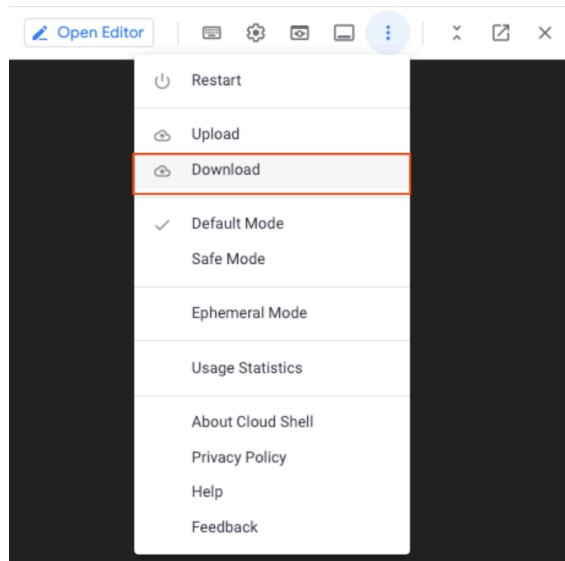
前往並執行 `setup_psc.sh` 指令碼，完成後會建立名為 **atlasEndpoints-psc-endpoint-us-west1.json** 的檔案。這個 JSON 檔案包含部署作業下一個步驟所需的 IP 位址和 Private Service Connect 端點名稱清單。

在 Cloud Shell 中執行下列操作：

```
sh setup_psc.sh
```

注意：執行殼層需要 5 到 7 分鐘，完成後就會部署消費者端點。前往「網路服務」→「Private Service Connect」→「已連線的端點」，查看 50 個端點的部署作業。

指令碼完成後，請使用 Cloud Shell 編輯器在本機下載 **atlasEndpoints-psc-endpoint-us-west1.json**。



Download File/Folder

File path

File paths are relative to home directory.

▼

- ▶
- ▶
- ▶
- ▶
- ▶
-
-
-

[CANCEL](#) [DOWNLOAD](#)

上傳 JSON 檔案

上傳先前儲存的 JSON 檔案 **atlasEndpoints-psc-endpoint-us-west1.json**。

Create a Private Endpoint M10 & ABOVE ONLY

✓ Cloud Provider

✓ Service Attachments

✓ Endpoints

✓ Set Up Endpoints

Finalize

Upload JSON File
Upload the outputted JSON file containing a list of IP addresses and Private Service Connect Endpoint Names

[Upload JSON File](#)

[Back](#) [Create](#)

×

Create a Private Endpoint M10 & ABOVE ONLY

✓ Cloud Provider

✓ Service Attachments

✓ Endpoints

✓ Set Up Endpoints

Finalize

Upload JSON File

Upload the outputted JSON file containing a list of IP addresses and Private Service Connect Endpoint Names

Upload JSON File

✓ atlasEndpoints-psc-endpoint-us-west1.json

Back

Create

選取「建立」

驗證 Private Service Connect 端點

在 MongoDB 使用者介面中，依序前往專案、安全性 → 網路存取權 → 私人端點。選取專屬叢集分頁後，端點轉換為可用狀態需要 10 分鐘。

可用狀態

Network Access

IP Access List

Peering

Private Endpoint

Dedicated Cluster

Serverless Instance

Federated Database Instance / Online Archive

+ ADD PRIVATE ENDPOINT

Cloud Provider	Region	Atlas Endpoint Service	Atlas Endpoint Service Status	Endpoint	Endpoint Status	Actions
GCP	us-west1	projects/p-npwmzbxmzmaejhj2ynla0g/regions/us-west1/serviceAttachments/so-us-west1-61485ec2ae9d2e48568bf84f-0-491	Available	psc-endpoint-us-west1	Available	EDIT TERMINATE

在 Google Cloud 控制台中，前往「網路服務」→「Private Service Connect」，然後選取「已連線的端點」分頁標籤，顯示從「待處理」轉換為「已接受」的消費者端點，如下所示：

Network services

Private Service Connect

CONNECTED ENDPOINTS

PUBLISHED SERVICES

Private Service Connect lets you connect privately and securely to Services. [Learn more](#)

Connections

50 in total

Accepted

26

Rejected

0

Pending

24

Closed

0

Endpoints

+ CONNECT ENDPOINT

Filter

Enter property name or value

☐	Endpoint	Status	PSC Connection ID	Target	Network	Region	IP address	Namespace
☐	psc-endpoint-us-west1-0	Accepted	69295173089626637	Published service	consumer-vpc	us-west1 (with global access)	192.168.10.13	goog-psc-default
☐	psc-endpoint-us-west1-1	Accepted	69295173089626638	Published service	consumer-vpc	us-west1 (with global access)	192.168.10.14	goog-psc-default
☐	psc-endpoint-us-west1-10	Accepted	69295173089626653	Published service	consumer-vpc	us-west1 (with global access)	192.168.10.29	goog-psc-default
☐	psc-endpoint-us-west1-11	Accepted	69295173089626654	Published service	consumer-vpc	us-west1 (with global access)	192.168.10.30	goog-psc-default
☐	psc-endpoint-us-west1-12	Accepted	69295173089626655	Published service	consumer-vpc	us-west1 (with global access)	192.168.10.31	goog-psc-default
☐	psc-endpoint-us-west1-13	Accepted	69295173089626656	Published service	consumer-vpc	us-west1 (with global access)	192.168.10.32	goog-psc-default
☐	psc-endpoint-us-west1-14	Accepted	69295173089626658	Published service	consumer-vpc	us-west1 (with global access)	192.168.10.34	goog-psc-default
☐	psc-endpoint-us-west1-15	Accepted	69295173089626659	Published service	consumer-vpc	us-west1 (with global access)	192.168.10.35	goog-psc-default
☐	psc-endpoint-us-west1-16	Accepted	69295173089626660	Published service	consumer-vpc	us-west1 (with global access)	192.168.10.36	goog-psc-default
☐	psc-endpoint-us-west1-17	Accepted	69295173089626661	Published service	consumer-vpc	us-west1 (with global access)	192.168.10.37	goog-psc-default
☐	psc-endpoint-us-west1-18	Accepted	69295173089626676	Published service	consumer-vpc	us-west1 (with global access)	192.168.10.52	goog-psc-default
☐	psc-endpoint-us-west1-19	Accepted	69295173089626677	Published service	consumer-vpc	us-west1 (with global access)	192.168.10.53	goog-psc-default
☐	psc-endpoint-us-west1-2	Accepted	69295173089626639	Published service	consumer-vpc	us-west1 (with global access)	192.168.10.15	goog-psc-default
☐	psc-endpoint-us-west1-20	Accepted	69295173089626678	Published service	consumer-vpc	us-west1 (with global access)	192.168.10.54	goog-psc-default
☐	psc-endpoint-us-west1-21	Accepted	69295173089626679	Published service	consumer-vpc	us-west1 (with global access)	192.168.10.55	goog-psc-default
☐	psc-endpoint-us-west1-22	Accepted	69295173089626680	Published service	consumer-vpc	us-west1 (with global access)	192.168.10.56	goog-psc-default
☐	psc-endpoint-us-west1-23	Accepted	69295173089626681	Published service	consumer-vpc	us-west1 (with global access)	192.168.10.57	goog-psc-default
☐	psc-endpoint-us-west1-24	Accepted	69295173089626682	Published service	consumer-vpc	us-west1 (with global access)	192.168.10.58	goog-psc-default
☐	psc-endpoint-us-west1-25	Accepted	69295173089626683	Published service	consumer-vpc	us-west1 (with global access)	192.168.10.59	goog-psc-default
☐	psc-endpoint-us-west1-26	Pending	69295173089626684	Published service	consumer-vpc	us-west1 (with global access)	192.168.10.60	goog-psc-default
☐	psc-endpoint-us-west1-27	Pending	69295173089626685	Published service	consumer-vpc	us-west1 (with global access)	192.168.10.61	goog-psc-default
☐	psc-endpoint-us-west1-28	Pending	69295173089626627	Published service	consumer-vpc	us-west1 (with global access)	192.168.10.3	goog-psc-default
☐	psc-endpoint-us-west1-29	Pending	69295173089626628	Published service	consumer-vpc	us-west1 (with global access)	192.168.10.4	goog-psc-default

注意：Private Service Connect 轉換作業最多可能需要 10 分鐘。

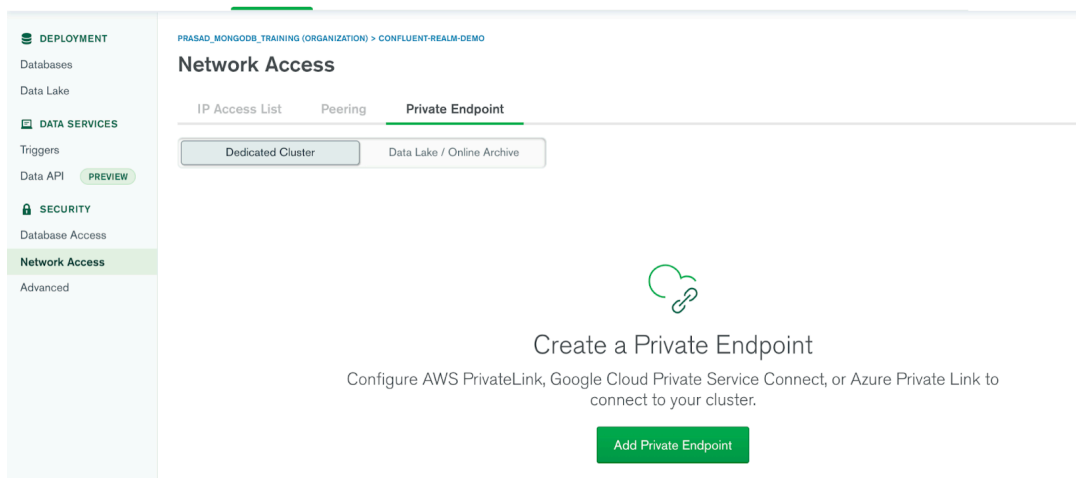
步驟 8 · 約 10 分鐘

8. 為 us-west2 建立私人端點

- 登入 Atlas 帳戶並前往專案。

在 MongoDB Atlas 中準備私人端點

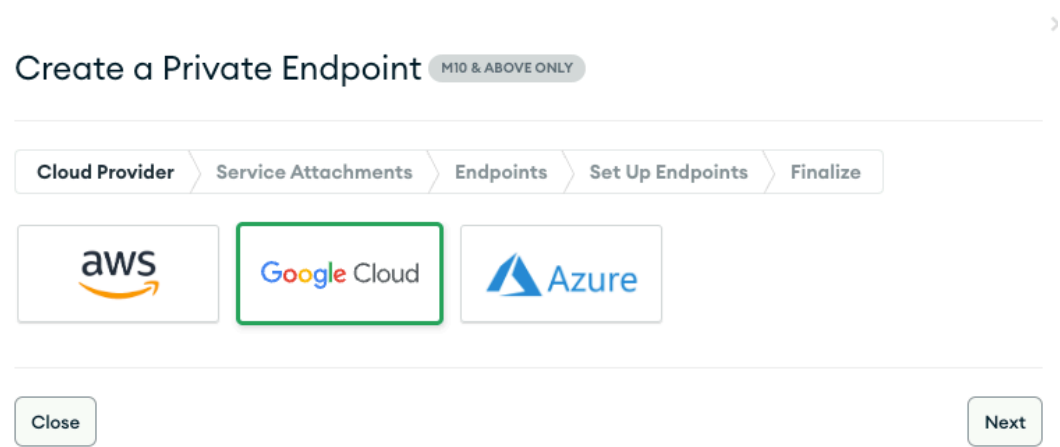
- 依序選取「網路存取」→「私人端點」→「專屬叢集」→「新增私人端點」



注意：每個專案的每個區域僅支援一個私人端點。如果 us-west1 和 us-west2 中有私人端點，請刪除叢集，然後在其他區域重新部署。

雲端服務供應商

- 選取 Google Cloud，然後按一下「下一步」



服務連結

- 選取 us-west2 地區，然後按一下「下一步」

Create a Private Endpoint M10 & ABOVE ONLY

✓ Cloud Provider

Service Attachments

Endpoints

Set Up Endpoints

Finalize

i The process of creating a private endpoint in Google Cloud includes the creation of [Private Service Connect endpoints](#) that will support growing Atlas Usage. This will enable you to connect to multiple Atlas clusters privately. [See documentation to learn more](#) 

Atlas Region

Select Atlas Region

★ Recommended region **i**



Los Angeles (us-west2) ★



Billing

Pay-as-you-go! After this step, you will be billed hourly for each Google Cloud Private Service Connect region at a rate of \$0.03 per GB.

Back

Next

端點

- 如要建立 Private Service Connect 端點，請提供下列資訊：
- Google Cloud 雲端專案 ID：選取「顯示操作說明」查看詳細資料
- 虛擬私有雲名稱：consumer-vpc
- 子網路名稱：psc-endpoint-us-west2
- Private Service Connect 端點前置字元：psc-endpoint-us-west2

Create a Private Endpoint M10 & ABOVE ONLY

✓ Cloud Provider

✓ Service Attachments

Endpoints

Set Up Endpoints

Finalize

🔄 **Creating Atlas Endpoint Service...** This might take 3-4 minutes to process

Once your Atlas Endpoint Service is ready, you will be able to create your Private Service Connect endpoints in your Google Cloud project using the [Google Cloud CLI](#). When you create Private Service Connect endpoints, you specify a subnet in your VPC network; you may either create a new subnet to encapsulate your endpoints or use an existing one. Enter the following information to populate the script in the next step.

Google Cloud Project ID
Find this in the Dashboard page in the Google Cloud Platform.
[Show instruction](#)

VPC Name
Find this on the VPC Networks page in the Google Cloud Platform.
[Show instruction](#)

Subnet Name ⓘ
Find this on the VPC Networks page in Google Cloud Platform. This subnet will need to have at least 50 unassigned IP addresses.
[Show instruction](#)

Private Service Connect Endpoint Prefix
Enter a unique name for your Private Endpoint. This will be the prefix for all forwarding rules created.

Back

Next

注意：端點建立作業需要 3 到 4 分鐘，完成後即可選取「下一步」

✓ Cloud Provider

✓ Service Attachments

Endpoints

Set Up Endpoints

Finalize

🔄 **Creating Atlas Endpoint Service...** This might take 3-4 minutes to process

設定端點

在下一節中，系統會產生名為 `setup_psc.sh` 的殼層指令碼，請將其儲存在本機。儲存後，請編輯殼層指令碼，允許 PSC 全域存取。您可以在 Google Cloud 雲端專案的 Cloud Shell 中執行這項操作。

注意：請先在 Cloud Shell 中移動並刪除先前建立的 `us-west1 setup_psc.sh` 和 `atlasEndpoints-psc-endpoint-us-west1.json`，再繼續操作。

- 範例殼層指令碼，輸出內容的值會有所不同

Create a Private Endpoint MIG & ABOVE ONLY

✓ Cloud Provider ✓ Service Attachments ✓ Endpoints **Set Up Endpoints** Finalize

Use the following generated script to set up your private endpoints. This script will reserve static internal IP addresses in the subnet specified in the previous step, and create forwarding rules to the Atlas Private Endpoint Service. For alternative options to set up your Private Service Connect endpoints, please review our [documentation](#).

1. Install gcloud command-line interface

You will need to have the gcloud command-line interface installed for the next steps. Follow the [instructions here](#) to install and run it on your local machine.

2. Copy and save shell script

Copy this script and save it to a file named `setup_psc.sh`.

```
#!/bin/bash
gcloud config set project [REDACTED]

for i in {0..49}
do
  gcloud compute addresses create psc-endpoint-us-west2-ip-$i --region=us-west2 --
  subnet=psc-endpoint-us-west2
done

for i in {0..49}
do
  if [ $(gcloud compute addresses describe psc-endpoint-us-west2-ip-$i --region=us
  -west2 --format="value(status)") != "RESERVED" ]; then
    echo "psc-endpoint-us-west2-ip-$i is not RESERVED";
    exit 1;
  fi
done

for i in {0..49}
do
  gcloud compute forwarding-rules create psc-endpoint-us-west2-$i --region=us-west
  2 --network=consumer-vpc --address=psc-endpoint-us-west2-ip-$i --target-service-at
  tachment=projects/p-npwszmzelxznmaejhj2vniq0q/regions/us-west2/serviceAttachments/s
  a-us-west2-61485ec2ae9d2e48568bf84f-$i
done

if [ $(gcloud compute forwarding-rules list --regions=us-west2 --format="csv(no-he
  ading)(name)" --filter="(name:psc-endpoint-us-west2*)" | wc -l) -gt 50 ]; then
  echo "Project has too many forwarding rules that match prefix psc-endpoint-us-we
  st2. Either delete the competing resources or choose another endpoint prefix."
  exit 2;
fi

gcloud compute forwarding-rules list --regions=us-west2 --format="json(IPAddress,n
  ame)" --filter="name:(psc-endpoint-us-west2*)" > atlasEndpoints-psc-endpoint-us-we
  st2.json
```

3. Run shell script

Run the script in your shell. Navigate to the directory where the script is saved and run it using the command below. This will output a file called `atlasEndpoints-psc-endpoint-us-west2.json` containing a list of IP addresses and forwarding rule names that you will need in the next step.

sh setup_psc.sh

Copy

Back

Next

- 從 MongoDB 控制台複製殼層指令碼，然後將內容儲存在 Google Cloud Cloud Shell 終端機中，並確保將指令碼儲存為 `setup_psc.sh`

更新前範例：

```
#!/bin/bash
gcloud config set project yourprojectname

for i in {0..49}
do
    gcloud compute addresses create psc-endpoint-us-west2-ip-$i --region=us-west2 --subnet=psc-
endpoint-us-west2
done

for i in {0..49}
do
    if [ $(gcloud compute addresses describe psc-endpoint-us-west2-ip-$i --region=us-west2 --
format="value(status)") != "RESERVED" ]; then
        echo "psc-endpoint-us-west2-ip-$i is not RESERVED";
        exit 1;
    fi
done

for i in {0..49}
do
    gcloud compute forwarding-rules create psc-endpoint-us-west2-$i --region=us-west2 --
network=consumer-vpc --address=psc-endpoint-us-west2-ip-$i --target-service-
attachment=projects/p-npwszmzelxznmaejhj2vn1q0q/regions/us-west2/serviceAttachments/sa-us-
west2-61485ec2ae9d2e48568bf84f-$i
done

if [ $(gcloud compute forwarding-rules list --regions=us-west2 --format="csv[no-heading]
(name)" --filter="(name:psc-endpoint-us-west2*)" | wc -l) -gt 50 ]; then
    echo "Project has too many forwarding rules that match prefix psc-endpoint-us-west2. Either
delete the competing resources or choose another endpoint prefix."
    exit 2;
fi

gcloud compute forwarding-rules list --regions=us-west2 --format="json(IPAddress,name)" --
filter="name:(psc-endpoint-us-west2*)" > atlasEndpoints-psc-endpoint-us-west2.json
```

更新殼層指令碼，支援全域存取

使用 nano 或 vi 編輯器找出並更新殼層指令碼，語法如下：

```
gcloud beta compute forwarding-rules create psc-endpoint-us-west2-$i --region=us-west2 --
network=consumer-vpc --address=psc-endpoint-us-west2-ip-$i --target-service-
attachment=projects/p-npwszmzelxznmaejhj2v
n1q0q/regions/us-west2/serviceAttachments/sa-us-west2-61485ec2ae9d2e48568bf84f-$i --allow-
psc-global-access
done
```

更新後的範例：

```
#!/bin/bash
gcloud config set project yourprojectname

for i in {0..49}
do
    gcloud compute addresses create psc-endpoint-us-west2-ip-$i --region=us-west2 --subnet=psc-endpoint-us-west2
done

for i in {0..49}
do
    if [ $(gcloud compute addresses describe psc-endpoint-us-west2-ip-$i --region=us-west2 --format="value(status)") != "RESERVED" ]; then
        echo "psc-endpoint-us-west2-ip-$i is not RESERVED";
        exit 1;
    fi
done

for i in {0..49}
do
    gcloud beta compute forwarding-rules create psc-endpoint-us-west2-$i --region=us-west2 --network=consumer-vpc --address=psc-endpoint-us-west2-ip-$i --target-service-attachment=projects/p-npwszmzelxznmaejhj2vnlq0q/regions/us-west2/serviceAttachments/sa-us-west2-61485ec2ae9d2e48568bf84f-$i --allow-psc-global-access
done

if [ $(gcloud compute forwarding-rules list --regions=us-west2 --format="csv[no-heading](name)" --filter="(name:psc-endpoint-us-west2*)" | wc -l) -gt 50 ]; then
    echo "Project has too many forwarding rules that match prefix psc-endpoint-us-west2. Either delete the competing resources or choose another endpoint prefix."
    exit 2;
fi

gcloud compute forwarding-rules list --regions=us-west2 --format="json(IPAddress,name)" --filter="name:(psc-endpoint-us-west2*)" > atlasEndpoints-psc-endpoint-us-west2.json
```

執行殼層指令碼

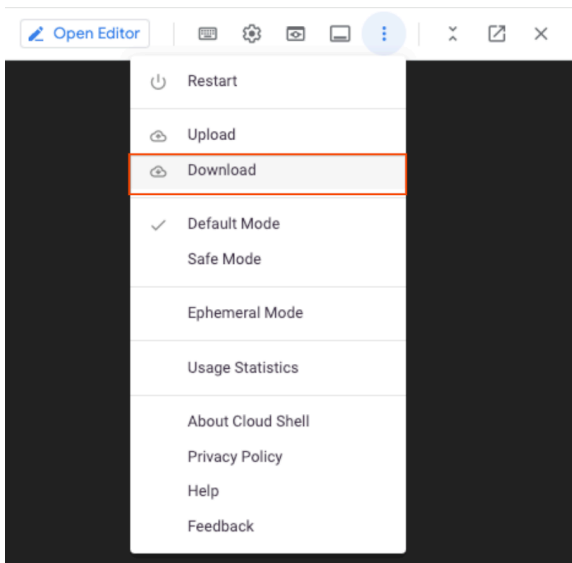
前往並執行 `setup_psc.sh` 指令碼，完成後會建立名為 **atlasEndpoints-psc-endpoint-us-west2.json** 的檔案。這個 JSON 檔案包含部署作業下一個步驟所需的 IP 位址和 Private Service Connect 端點名稱清單。

在 Cloud Shell 中執行下列操作：

```
sh setup_psc.sh
```

注意：執行殼層需要 5 到 7 分鐘，完成後就會部署消費者端點。前往「網路服務」→「Private Service Connect」→「已連線的端點」，查看 50 個端點的部署作業。

指令碼完成後，請使用 Cloud Shell 編輯器在本機下載 atlasEndpoints-psc-endpoint-us-west2.json 。



Download File/Folder

File path

File paths are relative to home directory.

▼

/home/username/

- ▶ bin
- ▶ devel
- ▶ file:
- ▶ GOPATH
- ▶ json

README-cloudshell.txt

atlasEndpoints-psc-endpoint-us-west2.json

codeLab

CANCEL DOWNLOAD

上傳 JSON 檔案

上傳先前儲存的 JSON 檔案 atlasEndpoints-psc-endpoint-us-west2.json

Create a Private Endpoint M10 & ABOVE ONLY

✔ Cloud Provider

✔ Service Attachments

✔ Endpoints

✔ Set Up Endpoints

Finalize

Upload JSON File

Upload the outputted JSON file containing a list of IP addresses and Private Service Connect Endpoint Names

Upload JSON File

Back

Create

Create a Private Endpoint M10 & ABOVE ONLY

✔ Cloud Provider

✔ Service Attachments

✔ Endpoints

✔ Set Up Endpoints

Finalize

Upload JSON File

Upload the outputted JSON file containing a list of IP addresses and Private Service Connect Endpoint Names

Upload JSON File

✔ atlasEndpoints-psc-endpoint-us-west1.json

Back

Create

選取「建立」

驗證 Private Service Connect 端點

在 MongoDB 使用者介面中，依序前往專案、安全性 → 網路存取權 → 私人端點。選取分頁後，專屬叢集的端點會在 10 分鐘後轉換為可用狀態。

可用狀態：

Network Access

[IP Access List](#)
[Peering](#)
[Private Endpoint](#)

Dedicated Cluster

Serverless Instance

Federated Database Instance / Online Archive

[+ ADD PRIVATE ENDPOINT](#)

Cloud Provider	Region	Atlas Endpoint Service	Atlas Endpoint Service Status	Endpoint	Endpoint Status	Actions
GCP	us-west1	projects/p-npwsrmzelznmajhj2vnlq0a/regions/us-west1/serviceAttachments/sa-us-west1-61485ec2ae9d2e48568bfb64f0-49f 	 Available	psc-endpoint-us-west1	 Available	EDIT TERMINATE
GCP	us-west2	projects/p-npwsrmzelznmajhj2vnlq0a/regions/us-west2/serviceAttachments/sa-us-west2-61485ec2ae9d2e48568bfb64f0-49f 	 Available	psc-endpoint-us-west2	 Available	EDIT TERMINATE

在 Google Cloud 控制台中，前往「網路服務」→「Private Service Connect」，然後選取「已連線的端點」分頁，顯示從「待處理」轉換為「已接受」的消費者端點，如下例所示。消費者端點共部署 100 個端點，必須先轉換為已接受，才能進行下一個步驟。

Network services

Load balancing

Cloud DNS

Cloud CDN

Cloud NAT

Traffic Director

Service Directory

Cloud Domains

Private Service Connect

Service Observer

Private Service Connect

CONNECTED ENDPOINTS
PUBLISHED SERVICES

Private Service Connect lets you connect privately and securely to Services. [Learn more](#)

Connections
100 in total

Accepted
52

Rejected
0

Pending
48

Closed
0

Endpoints
+ CONNECT ENDPOINT

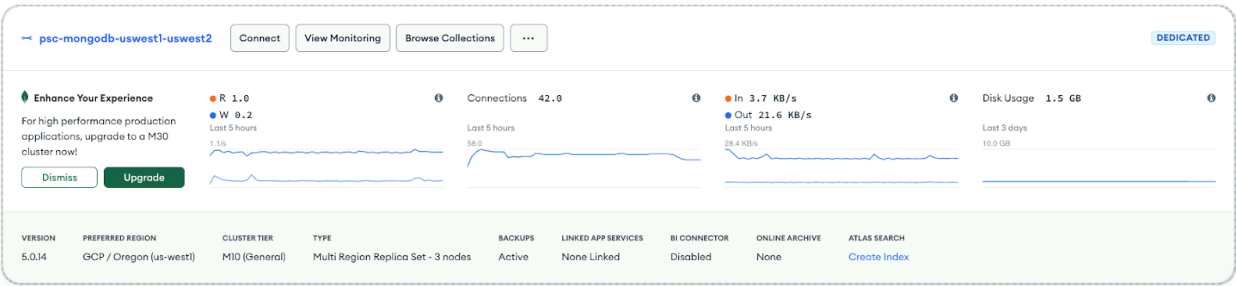
Filter
us-west2
Enter property name or value

☐	Endpoint ↑	Status	PSC Connection ID	Target	Network	Region	IP address	Namespace	
☐	psc-endpoint-us-west2-0	Accepted	69295172744120834	Published service	consumer-vpc	us-west2 (with global access)	172.16.10.2	goog-psc-default	⋮
☐	psc-endpoint-us-west2-1	Accepted	69295172744120835	Published service	consumer-vpc	us-west2 (with global access)	172.16.10.3	goog-psc-default	⋮
☐	psc-endpoint-us-west2-10	Pending	69295172744120844	Published service	consumer-vpc	us-west2 (with global access)	172.16.10.12	goog-psc-default	⋮
☐	psc-endpoint-us-west2-11	Pending	69295172744120845	Published service	consumer-vpc	us-west2 (with global access)	172.16.10.13	goog-psc-default	⋮
☐	psc-endpoint-us-west2-12	Pending	69295172744120846	Published service	consumer-vpc	us-west2 (with global access)	172.16.10.14	goog-psc-default	⋮
☐	psc-endpoint-us-west2-13	Pending	69295172744120847	Published service	consumer-vpc	us-west2 (with global access)	172.16.10.15	goog-psc-default	⋮
☐	psc-endpoint-us-west2-14	Pending	69295172744120848	Published service	consumer-vpc	us-west2 (with global access)	172.16.10.16	goog-psc-default	⋮
☐	psc-endpoint-us-west2-15	Pending	69295172744120849	Published service	consumer-vpc	us-west2 (with global access)	172.16.10.17	goog-psc-default	⋮

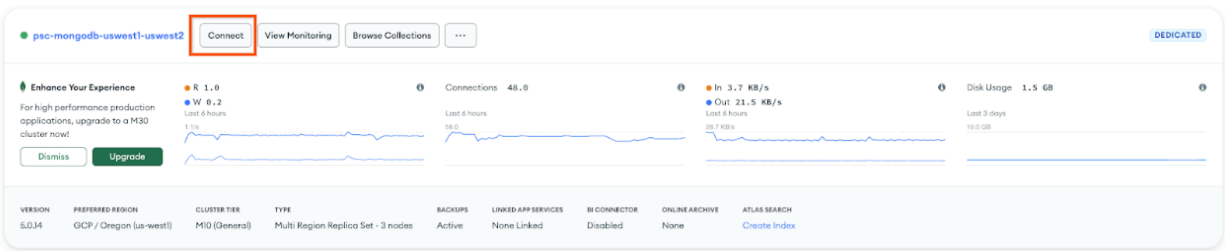
步驟 9 · 約 15 分鐘

9. 從私人端點連線至 MongoDB Atlas

Private Service Connect 連線獲得接受後，更新 MongoDB 叢集還需要額外時間 (10 到 15 分鐘)。在 MongoDB UI 中，灰色外框代表叢集更新，因此無法連線至私人端點。



找出部署作業並選取「連線」(請注意，灰色方塊已不再顯示)



選擇連線類型 → 私人端點，然後選取「選擇連線方法」

Connect to psc-mongodb-uswest1-uswest2

Setup connection security

Choose a connection method

Connect

You need to secure your MongoDB Atlas cluster before you can use it. Set which users, IP addresses, or private endpoints can access your cluster now. [Read more](#)

Choose connection type

Private Endpoint

Standard Connection

Choose private endpoint

Multi-region clusters only have a single private endpoint.

Add a connection IP address

All servers with access to your endpoints can connect to your cluster without explicit IP access configuration. If you are having trouble connecting with your Private Endpoint, please visit our [troubleshoot documentation](#)

Create a database user

A MongoDB user has been added to this project. Not yours? Create one in the [MongoDB Users tab](#).

You'll need your MongoDB user's credentials in the next step.

Close

Choose a connection method

選取「Connect with the MongoDB Shell」(使用 MongoDB Shell 連線)

Connect to psc-mongodb-uswest1-uswest2

✓ Setup connection security Choose a connection method Connect

Choose a connection method [View documentation](#)

Get your pre-formatted connection string by selecting your tool below.



Connect with the MongoDB Shell
Interact with your cluster using MongoDB's interactive Javascript interface

>



Connect your application
Connect your application to your cluster using MongoDB's native drivers

>



Connect using MongoDB Compass
Explore, modify, and visualize your data with MongoDB's GUI

>



Connect using VS Code
Connect to a MongoDB host in Visual Studio Code

>



Connect Your Business Intelligence Tool ([Enable the BI Connector](#))
Get connection settings for Business Intelligence tools and visualize your data

>

Go Back Close

選取「I do not have the MongoDB Shell installed, Ubuntu 20.4」，並務必將步驟 1 和步驟 3 的內容複製到記事本。

Connect to psc-mongodb-uswest1-uswest2

✓ Setup connection security ✓ Choose a connection method Connect

I do not have the MongoDB Shell installed I have the MongoDB Shell installed

1 Select your operating system and download the mongosh

Ubuntu 20.04

Download mongosh (1.7.1) or Copy download URL

2 Add <your mongosh's download directory>/bin to your \$PATH variable. [How to](#)

3 Run your connection string in your command line

Use this connection string in your application:

```
mongosh "mongodb+srv://psc-mongodb-uswest1-uswest2-pl-0.2wqno.mongodb.net/myFirstDatabase?--apiVersion 1 --username <username>"
```

Replace **myFirstDatabase** with the name of the database that connections will use by default. You will be prompted for the password for the Database User, **<username>**. When entering your password, make sure all special characters are [URL encoded](#).

Having trouble connecting? [View our troubleshooting documentation](#)

Go Back Close

步驟 10 · 約 10 分鐘

10. 安裝 mongosh 應用程式

安裝前，您需要根據先前在步驟 1 和 3 中複製的值建立指令字串。接著，您會使用 Cloud Shell 透過 SSH 連線至 vm1，然後安裝並驗證主要 (us-west1) 資料庫的 mongosh 應用程式。在 consumer-vpc 中建立 vm1 時，已安裝 Ubuntu 20.4 映像檔。

選擇連線方法：步驟 1，複製下載網址

1 Select your operating system and download the mongosh

Ubuntu 20.04

Download mongosh (1.7.1)

Copy download URL

指令字串範例，請替換為自訂值：

```
https://downloads.mongodb.com/compass/mongodb-mongosh_1.7.1_amd64.deb
```

選擇連線方式 (步驟 3)。

3 Run your connection string in your command line

Use this connection string in your application:

```
mongosh "mongodb+srv://psc-mongodb-uswest1-uswest2-pl-0.2wqno.mongodb.net/myFirstDatabase" --apiVersion 1 --username <username>
```

Replace **myFirstDatabase** with the name of the database that connections will use by default. You will be prompted for the password for the Database User, **<username>**. When entering your password, make sure all special characters are **URL encoded**.

指令字串範例，請替換為自訂值：

```
mongosh "mongodb+srv://psc-mongodb-uswest1-uswest2-pl-0.2wqno.mongodb.net/psc-mongodb-uswest1-uswest2" --apiVersion 1 --username codelab
```

登入 vm1

在 Cloud Shell 中執行下列操作：

```
gcloud config list project
gcloud config set project [YOUR-PROJECT-NAME]
projectname=YOUR-PROJECT-NAME
echo $projectname
```

在 Cloud Shell 中使用 IAP 登入 vm1，如果逾時請重試。

```
gcloud compute ssh vm1 --project=$projectname --zone=us-west1-a --tunnel-through-iap
```


從作業系統執行安裝作業

從 Cloud Shell OS 登入執行安裝作業 ([詳情請參閱這篇文章](#))，並使用自訂字串更新下方語法

```
wget -qO - https://www.mongodb.org/static/pgp/server-6.0.asc | sudo apt-key add -
```

作業應會傳回

OK

。

```
echo "deb [ arch=amd64,arm64 ] https://repo.mongodb.org/apt/ubuntu focal/mongodb-org/6.0 multiverse" | sudo tee /etc/apt/sources.list.d/mongodb-org-6.0.list
```

```
sudo apt-get update -y
```

```
wget https://downloads.mongodb.com/compass/mongodb-mongosh_1.7.1_amd64.deb
```

```
sudo dpkg -i mongodb-mongosh_1.7.1_amd64.deb
```

連線至 MongoDB 部署

在 Cloud Shell OS 登入中執行下列操作。設定的使用者名稱和密碼為「codelab」。

```
mongosh
```

```
mongosh "mongodb+srv://psc-mongodb-uswest1-uswest2-pl-0.2wqno.mongodb.net/psc-mongodb-uswest1-uswest2" --apiVersion 1 --username codelab
```

範例如下：

對資料庫執行指令

在 Cloud Shell OS 登入中執行下列操作。

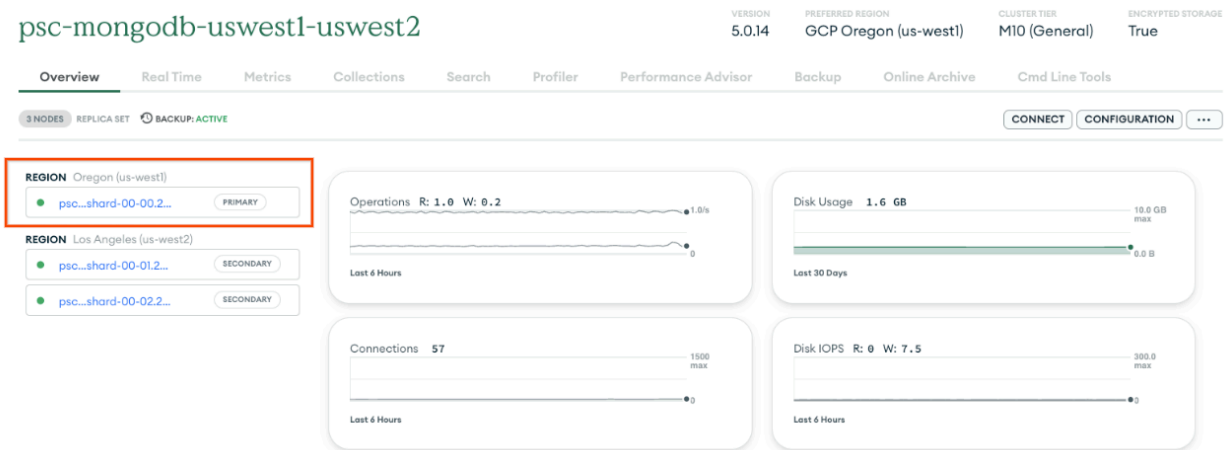
```
show dbs
use Company
db.Employee.insertOne({ "Name": "cosmo", "dept": "devops" })
db.Employee.findOne({ "Name": "cosmo" })
```

步驟 11 · 約 10 分鐘

11. 容錯移轉使用中的 MongoDB 區域 (us-west1)

執行容錯移轉前，請先驗證 us-west1 是主要節點，且 us-west2 有兩個次要節點。

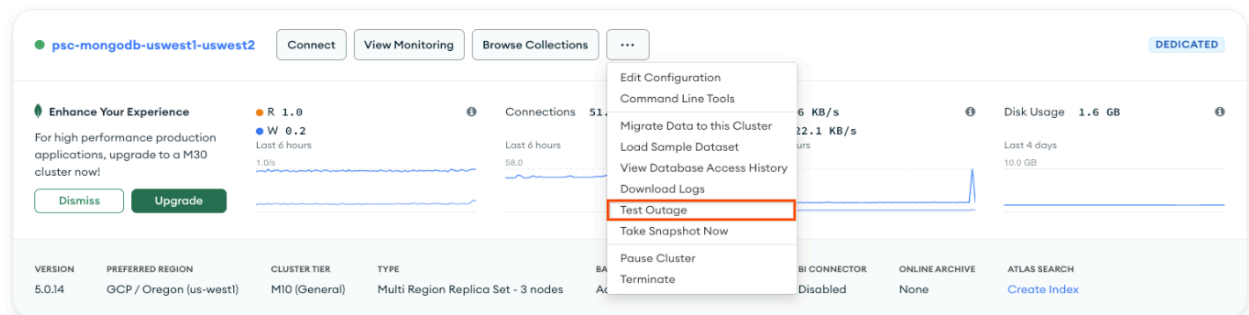
依序前往「Database」→「psc-mongodb-uswest1-uswest2」→「Overview」



在下一節中，請登入位於 us-west1 的 vm1，將主要 MongoDB 叢集區域 us-west1 容錯移轉，並確認資料庫仍可從 us-west2 的 MongoDB 叢集存取。

您可以透過 Atlas UI 測試主要和區域性容錯移轉。

- 登入 Atlas UI。
- 依序點選叢集名稱旁的 [...] 和「Test Outage」(測試中斷)。



- 選取「區域性服務中斷」→ 選取區域。

Test Outage

1

2

3

4

Choose Outage Scenario

Select Regions

Run and Review

End Simulation

Primary Failover

Regional Outage

This test simulates what happens when some deployed cluster node regions suffer an outage. Here's what to expect when you [test regional outage](#) :

1

2

3

4

Select Regions

Simulate Outage

Review Outage

End Simulation

✓

Region 1

✓

Region 1

Region 2

⚠

Region 1

●

Region 1

●

Region 2

?

Region 1

?

Region 1

●

Region 2

●

Region 1

●

Region 1

●

Region 2

Select regions you want to simulate an outage in.

All nodes in selected regions will be shut down.

Review whether your cluster is able to handle the regional outage.

After testing, you can end the simulation and restore connectivity.

Cancel

Select Regions

- 選取主要區域 us-west1 → 模擬區域中斷。

Test Outage

✓

2

3

4

Choose Outage Scenario

Select Regions

Run and Review

End Simulation

Select Regions to Simulate an Outage In

When a region is shut down, all electable, analytics and read-only nodes in the region will be unreachable.

	Provider	Region	Electable Nodes	Analytics Nodes	Read-Only Nodes
☒	GCP	Oregon (us-west1)	1	0	0
☐	GCP	Los Angeles (us-west2)	2	0	0

Total: 1/3 Electable Nodes

?

This test will leave the cluster with a majority quorum. There will be a primary and reads will succeed.

Go Back

Close

Simulate Regional Outage

選取叢集後，系統會在 3 到 4 分鐘內顯示中斷模擬結果

關閉視窗



Test Outage

✓

✓

3

4

Choose Outage Scenario

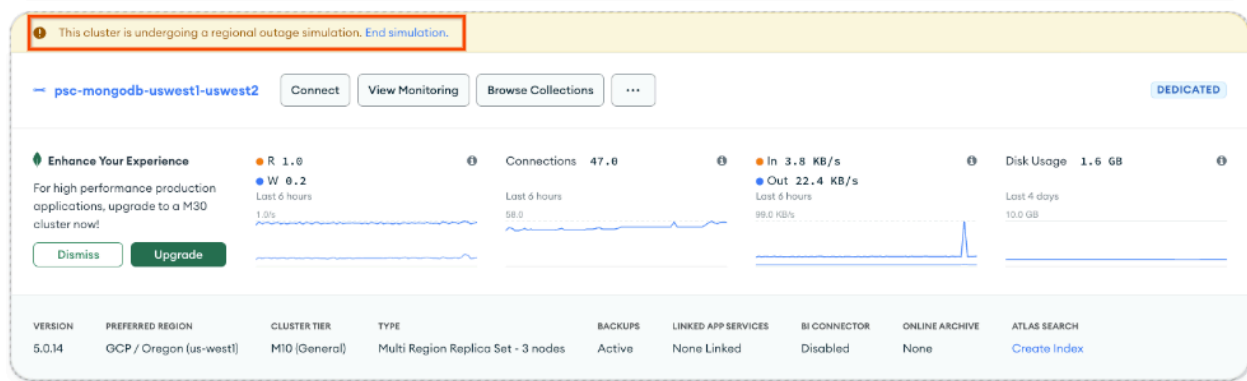
Select Regions

Run and Review

End Simulation

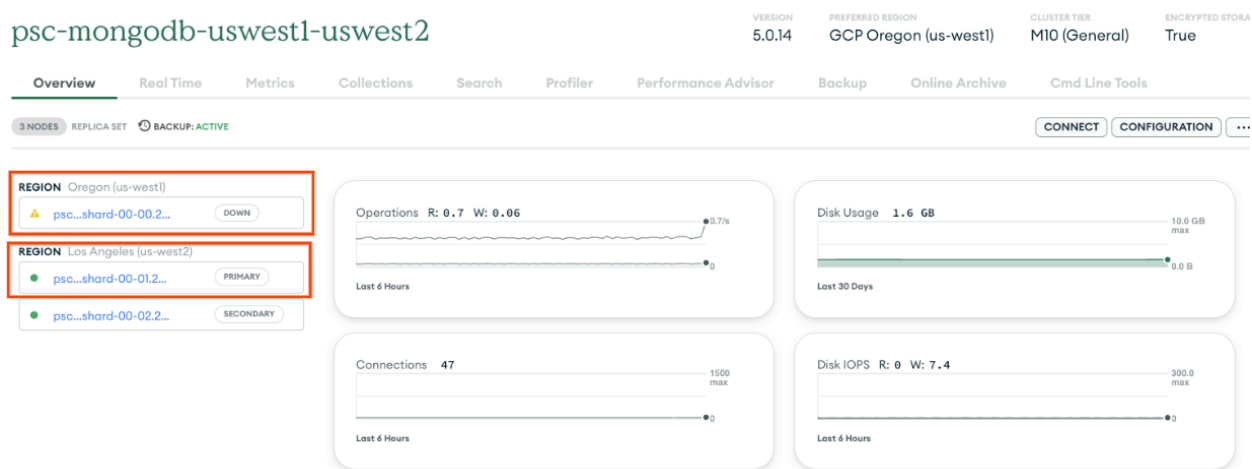
Simulating Regional Outage...

Atlas is removing connectivity from this cluster's nodes in the selected regions.



確認 us-west1 已停止運作，且 us-west2 現在已接管主要角色

依序前往「Database」→「psc-mongodb-uswest1-uswest2」→「Overview」



驗證與新主叢集 (us-west2) 的連線

登入位於 us-west1 的 vm1，並存取 us-west2 中的 mongodb，驗證 Private Service Connect 的全域存取權。

如果 Cloud Shell 工作階段已終止，請執行下列操作：

在 Cloud Shell 中執行下列操作：

```
gcloud config list project
gcloud config set project [YOUR-PROJECT-NAME]
projectname=YOUR-PROJECT-NAME
echo $projectname
```

在 Cloud Shell 中使用 IAP 登入 vm1，如果逾時請重試。

```
gcloud compute ssh vm1 --project=$projectname --zone=us-west1-a --tunnel-through-iap
```

連線至 MongoDB 部署

在 Cloud Shell OS 登入中執行下列操作。

```
mongosh
```

```
mongosh "mongodb+srv://psc-mongodb-uswest1-uswest2-pl-0.2wqno.mongodb.net/psc-mongodb-uswest1-uswest2" --apiVersion 1 --username codelab
```

範例如下：

```
google_com@vml:~$ mongosh
Current Mongosh Log ID: 63fd71d390d85b4ac9a85cce
Connecting to: mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+1.7.1
MongoNetworkError: connect ECONNREFUSED 127.0.0.1:27017
google_com@vml:~$
google_com@vml:~$ mongosh "mongodb+srv://psc-mongodb-uswest1-uswest2-pl-0.2wqno.mongodb.net/psc-mongodb-uswest1-uswest2" --apiVersion 1 --username codelab
Enter password: *****
Current Mongosh Log ID: 63fd723600dc17d4a294500d
Connecting to: mongodb+srv://<credentials>@psc-mongodb-uswest1-uswest2-pl-0.2wqno.mongodb.net/psc-mongodb-uswest1-uswest2?appName=mongosh+1.7.1
Using MongoDB: 5.0.14 (API Version 1)
Using Mongosh: 1.7.1

For mongosh info see: https://docs.mongodb.com/mongodb-shell/

To help improve our products, anonymous usage data is collected and sent to MongoDB periodically (https://www.mongodb.com/legal/privacy-policy).
You can opt-out by running the disableTelemetry() command.

Atlas atlas-11z04y-shard-0 [primary] psc-mongodb-uswest1-uswest2>
```

對資料庫執行指令

在 Cloud Shell OS 登入中執行下列操作。

```
show dbs
use Company
db.Employee.insertOne({ "Name": "cosmo", "dept": "devops" })
db.Employee.findOne()

exit
```

```
Atlas atlas-11z04y-shard-0 [primary] Company> show dbs
Company 40.00 KiB
admin 268.00 KiB
config 224.00 KiB
local 964.00 KiB
Atlas atlas-11z04y-shard-0 [primary] Company> use Company
already on db Company
Atlas atlas-11z04y-shard-0 [primary] Company> db.Employee.insertOne({ "Name": "cosmo", "dept": "devops" })
{ acknowledged: true, insertedId: ObjectId("63ff823ff7211778050af78a") }
Atlas atlas-11z04y-shard-0 [primary] Company> db.Employee.findOne()
{ _id: ObjectId("63ff7ecdf7211778050af789"), Name: 'cosmo', dept: 'devops' }
Atlas atlas-11z04y-shard-0 [primary] Company>
```

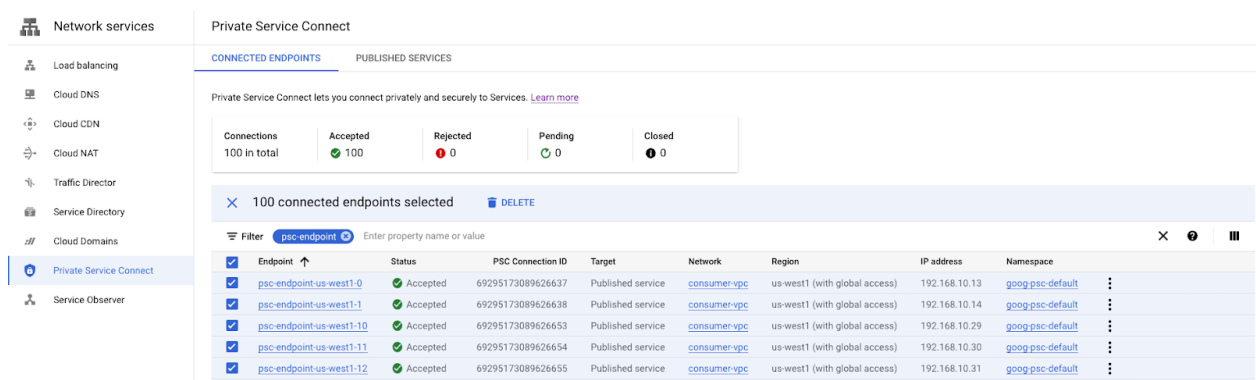
成功：您已驗證 PSC 全域存取權可讓消費者端點跨區域順暢連線，有助於提高可用性或因應區域性中斷。在本程式碼實驗室中，MongoDB 地區容錯轉移發生在位於 us-west1 的主要節點，因此次要區域 us-west2 接管了主要節點。雖然叢集導致區域服務中斷，但位於 us-west1 的消費者 VM1 仍成功連上 us-west2 的新主要叢集。

12. 清除

從 Cloud 控制台刪除消費者端點

依序前往「網路服務」→「Private Service Connect」→「已連線的端點」

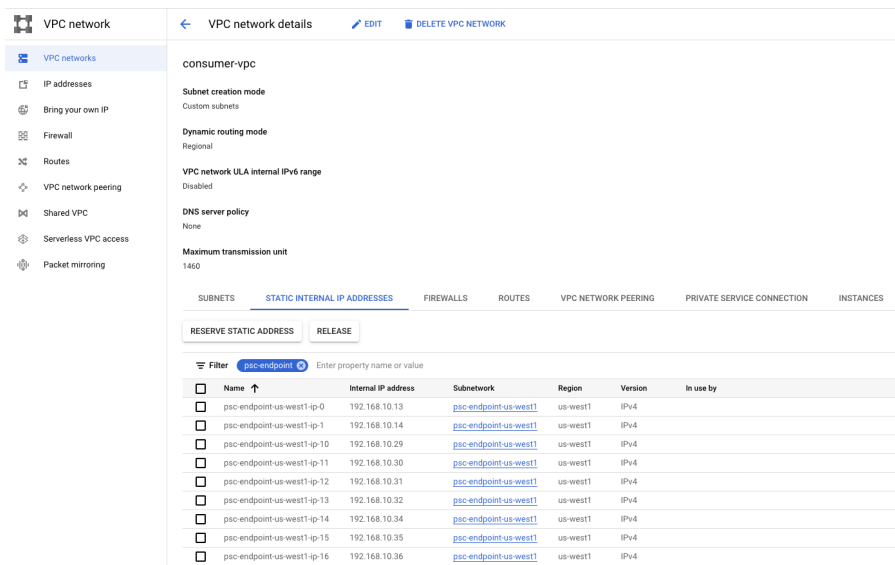
使用 **psc-endpoint** 篩選器，避免刪除非實驗室的消費者端點。選取所有端點 → 刪除



刪除與消費者端點相關聯的靜態內部 IP 位址

前往「VPC network」(虛擬私有雲網路) → 「consumer-vpc」 → 「STATIC INTERNAL IP ADDRESSES」(靜態內部 IP 位址)

使用 **psc-endpoint** 篩選器，排除可能刪除的非實驗室消費者端點，並將每頁列數增加至 100。選取所有端點 → 「發布」



從 Cloud Shell 刪除 Codelab 元件。

```

gcloud compute instances delete vm1 --zone=us-west1-a --quiet

gcloud compute networks subnets delete psc-endpoint-us-west1 vm-subnet --region=us-west1 --quiet

gcloud compute networks subnets delete psc-endpoint-us-west2 --region=us-west2 --quiet

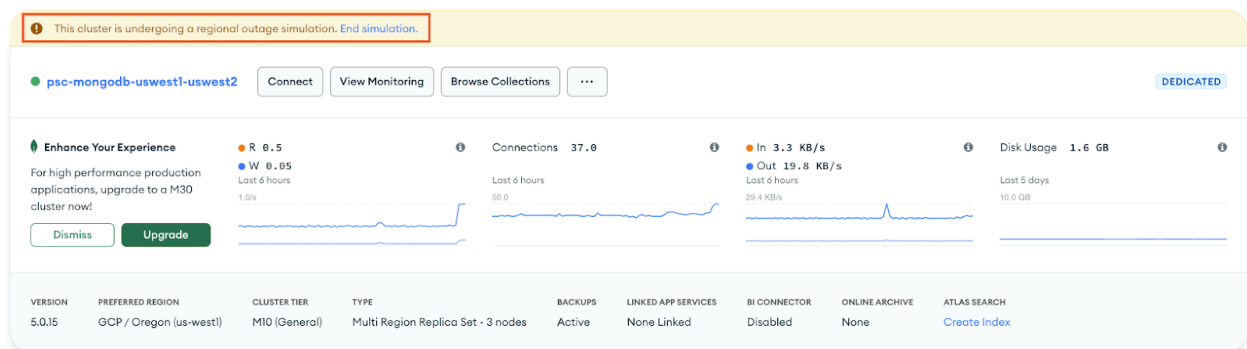
gcloud compute firewall-rules delete ssh-iap-consumer --quiet

gcloud compute routers delete consumer-cr --region=us-west1 --quiet

gcloud compute networks delete consumer-vpc --quiet

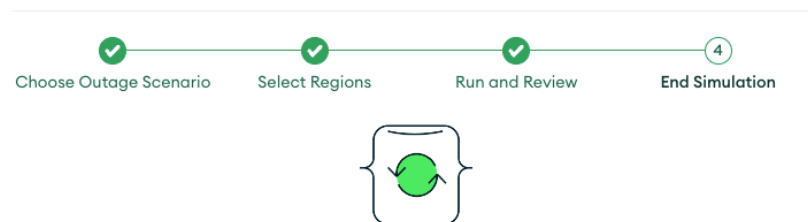
```

在 Atlas UI 中找出叢集 psc-mongodb-uswest1-uswest2 → 結束模擬



選取「結束中斷模擬」→「結束」

Test Outage



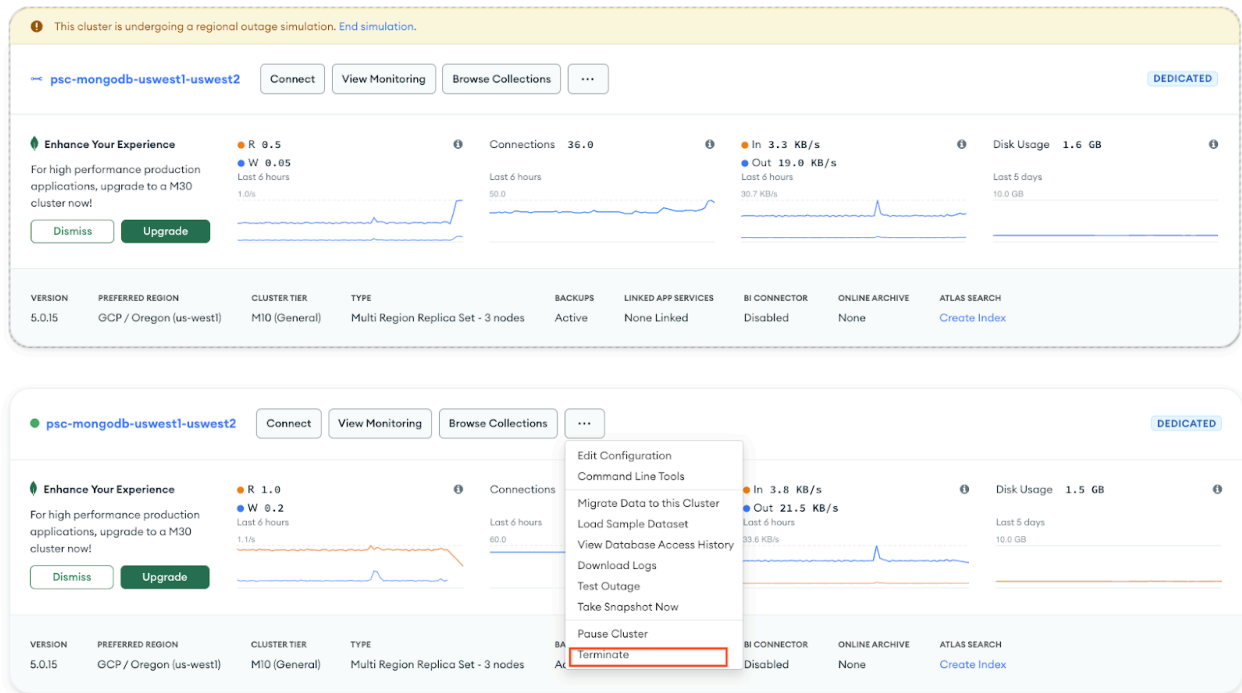
Regional Outage Simulation is Underway

Atlas has removed connectivity from the selected regions. Check your application to ensure reads and writes are still working as expected. Once you have finished testing, you can end the regional outage simulation.

Close

End Outage Simulation

叢集現在正在將 us-west1 還原為主要叢集，這項程序需要 3 到 4 分鐘。完成後，終止叢集，並注意灰色外框，表示狀態已變更。



插入叢集名稱 → 終止

Terminate Cluster: psc-mongodb-uswest1-uswest2

Note: This operation will delete the cluster and all of its data. It is **irreversible**. Backup is currently active for this cluster. Terminating your cluster will delete all of your backup snapshots immediately. If you want to keep old versions of your data, visit the [backup dashboard](#) and download your snapshots. Data Lake Datasets and Federated Database Instances will not be deleted.

Are you sure you want to terminate
PRASAD_MONGODB_TRAINING (Organization) > GCP-PoV-Cluster > **psc-mongodb-uswest1-uswest2**?

Please enter the cluster name to continue.

psc-mongodb-uswest1-uswest2

Cancel

Terminate

刪除與 us-west1 和 us-west2 相關聯的私人端點

在 Atlas UI 中，依序前往「Security」→「Network Access」→「Private Endpoint」，然後選取「Terminate」

Network Access

IP Access List						
Peering						
Private Endpoint						
Dedicated Cluster						
Serverless Instance						
Federated Database Instance / Online Archive						
+ ADD PRIVATE ENDPOINT						
Cloud Provider	Region	Atlas Endpoint Service	Atlas Endpoint Service Status	Endpoint	Endpoint Status	Actions
GCP	us-west1	projects/p-npwszmzelxzmoejhj2vn1q0a/regions/us-west1/serviceAttachments/sa-us-west1-61485ec2ae9d2e48568bf84f-[0-49]	Available	psc-endpoint-us-west1	Available	EDIT TERMINATE
GCP	us-west2	projects/p-npwszmzelxzmoejhj2vn1q0a/regions/us-west2/serviceAttachments/sa-us-west2-61485ec2ae9d2e48568bf84f-[0-49]	Available	psc-endpoint-us-west2	Available	EDIT TERMINATE

13. 恭喜

恭喜！您已成功設定並驗證 Private Service Connect 端點，可跨區域存取 MongoDB。您已成功建立消費端虛擬私有雲、多區域 MongoDB 和消費端端點。位於 us-west1 的 VM 在區域容錯移轉後，已成功連線至 us-west1 和 us-west2 中的 MongoDB。

Cosmopup 認為程式碼研究室很棒！



後續步驟

查看一些程式碼研究室...

- [使用 Private Service Connect 發布及使用 GKE 服務](#)
- [使用 Private Service Connect 發布及使用服務](#)
- [透過 Private Service Connect 和內部 TCP Proxy 負載平衡器，經由混合式網路連線至地端部署服務](#)
- [使用具備自動 DNS 設定的 Private Service Connect](#)

延伸閱讀和影片

- [Private Service Connect 總覽](#)
- [什麼是 Private Service Connect？](#)
- [支援的負載平衡器類型](#)

參考文件

- [內部 HTTP\(S\) 負載平衡器總覽](#)
 - [健康檢查總覽](#)
 - [如何使用 Private Service Connect 發布服務](#)
 - [Private Service Connect - 自動 DNS](#)
-